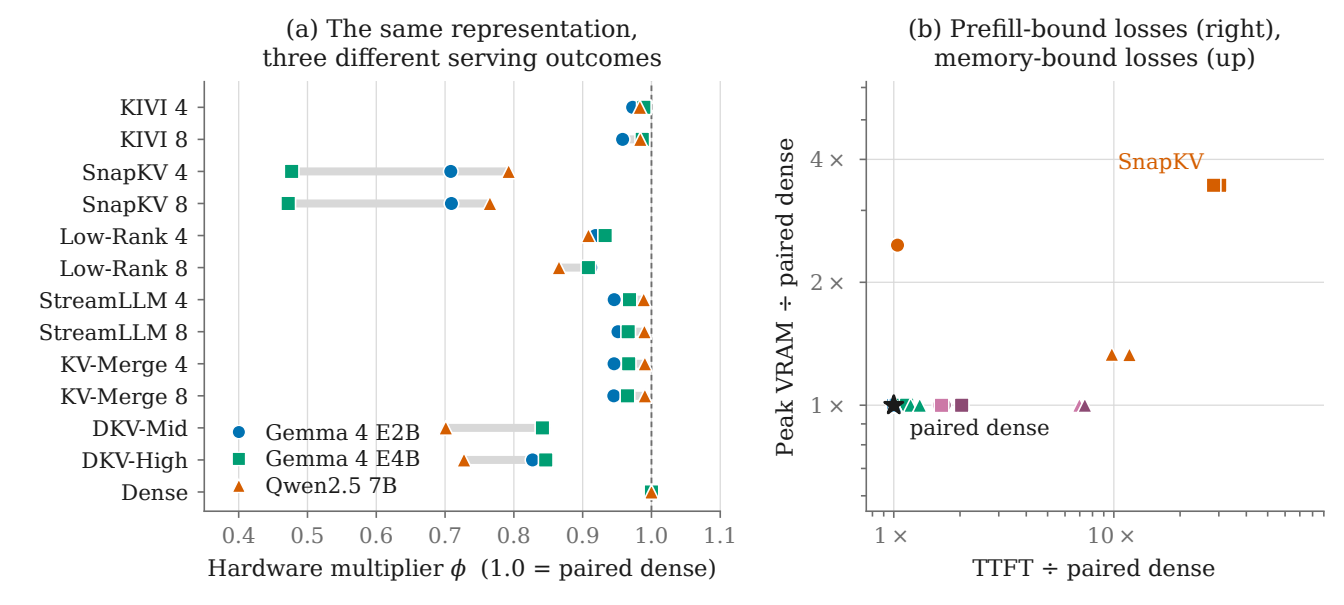
(a) The same representation,
three different serving outcomes
KIVI 4
KIVI 8
SnapKV 4
SnapKV 8
Low-Rank 4
Low-Rank 8
StreamLLM 4
StreamLLM 8
KV-Merge 4
KV-Merge 8
DKV-Mid
DKV-High
Dense
Gemma 4 E2B
Gemma 4 E4B
Qwen2.5 7B
0.4
0.5
0.6
0.7
0.8
0.9
1.0
1.1
Hardware multiplier ϕ (1.0 = paired dense)
(b) Prefill-bound losses (right),
memory-bound losses (up)
Peak VRAM ÷ paired dense
4 ×
2 ×
1 ×
SnapKV
paired dense
1 ×
10 ×
TTFT ÷ paired dense